EL NIÑO ACTION BRIEFING · 14 OCTOBER 2026

North (CINOR), Réunion

Not an El Niño drought: a record carried-over water deficit, leaking networks, heat, mosquito-borne disease bred in stored water, leptospirosis, and cyclones that become water outages, in a hot El Niño year. Here, the first job is to keep stored water covered and the Rivière Saint-Denis intake running through a cyclone.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
All Réunion (one region); figure for the Saint-Denis arrondissement	2 (Réunion: 5)	6-month, Sep 2026 – Feb 2027	Water outages after cyclones; dengue bred in stored water

16/16 recommendations checked. 0 confirmed as written, 9 corrected, 0 unverified; 7 contraindicated (the *Do not* cards). **The funding target fell 55%**, from the panel's €1,380,500 to €627,500: covers cut to one per household, and trigger cash moved to the island line RE-F01, not counted here. Exposure: Climate Impact Lab, *451,000 at Risk*, Sep 2026, pre-peer-review, heat only; its own published row for the arrondissement of Saint-Denis, which is exactly CINOR. Not our scaling.

EXECUTIVE SUMMARY — AT A GLANCE



20.90°S 55.50°E, France · IGN ADMIN EXPRESS, OpenStreetMap contributors; Natural Earth

- **The risk.** Garance (Feb 2025) killed 3 people in Saint-Denis and cut its upper quarters' water for 4+ days; stored water can breed dengue, which is sporadic now but with DENV2 back at 30% of typed cases.
- **First action.** The three communes, with neighbourhood associations, run breeding-site and water-cover rounds in the 16 priority neighbourhoods from November.
- **The money.** €279k / €628k / €1.14M (floor / target / ceiling). Lists, kits, and transport for rough-sleeper outreach save the most lives per euro (€60k).
- **Do not.** Never tell people to "store water" without "covered, emptied weekly": each rationing notice otherwise multiplies breeding sites.

What to do	**9** actions	p. 2
Who does what	RACI chart	p. 11
Where money does the most	funding	p. 11
Do not do this	**7** items	p. 14
What has happened here	the record	p. 16
How this was produced	methodology	p. 19
References	sources	p. 21

* Suggested formulation: check with a health worker before sharing.

THE PLACE AND THE SEASON

SCOPE

CINOR (Communauté intercommunale du Nord de La Réunion) is Saint-Denis, Sainte-Marie, and Sainte-Suzanne: 218,278 people (INSEE, populations légales 2023, in force 2026), 92,833 households, and a 34.2% poverty rate (INSEE, 2023). CINOR has held the drinking-water competence since 1 January 2020. The operators differ by commune: SEMOP Dionéo in Saint-Denis, CISE Réunion in Sainte-Marie, and Runéo in Sainte-Suzanne. This briefing is the first of six for Réunion and is written to be adopted as an annex to CINOR's intercommunal safeguard plan (PICS), *volet pénurie d'eau, chaleur et épidémies*. Each action carries its cyclone phase (avant, pendant, or sauvegarde).

COMMUNE	WATER OPERATOR	VIGIEAU, DRINKING WATER (14 OCT 2026)	NETWORK EFFICIENCY (2021)
Saint-Denis	SEMOP Dionéo (since 2023)	Alerte renforcée	66% (Grenelle target 78%)
Sainte-Marie	CISE Réunion	Alerte	61% (target 73%)
Sainte-Suzanne	Runéo	Alerte	78% (target 73%, met)

VigiEau shows the restriction level, not the state of supply. Prefectoral order of 9 October 2026, in force 10 October. Efficiency: Office de l'eau, *Chroniques de l'eau* n°133 (2023), 2021 data; island average 63%. Météo-France expects El Niño to bring Réunion rain that is on average in surplus, a clearly hotter season (about 19 days above 31°C at Gillot, in Sainte-Marie, against a normal of 2), and no change in cyclone exposure. The deficit comes from before this El Niño: 2025–26 was the driest rainy season in 54 years of records (−45%, provisional; −60 to −70% in the hauts), the second deficit season in a row after −30% in 2024–25 (Météo-France, provisional season review, 29 April 2026).

WHAT THIS PLACE SHOULD DO

RANKED

Ranked by expected lives saved per unit of effort before the peak. Backing lenses are shown per action; an action several lenses reached

* Suggested formulation: check with a health worker before sharing.

independently is the stronger bet.

1. From November, run breeding-site and water-cover rounds door to door in the 16 priority neighbourhoods, and print "couvrez vos réserves" on every CINOR water-cut notice

PARTIALLY CONFIRMED | PHASE: AVANT | PUBLIC HEALTH · SKEPTIC · FRONTLINE · HISTORIAN · PHILANTHROPIST · EMERGENCY MANAGER | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
The three communes, with neighbourhood associations; the ARS consulted on targeting	Rounds 2 Nov – 15 Dec 2026, then monthly	€208,000 (target): covers for 16,000 households at €13 delivered; floor a 1,000-household pilot, €13,000 (F03)	Dengue bred in water stored during cuts

CINOR has 16 priority neighbourhoods (QPV, décret 2024-1212): 12 in Saint-Denis (including Le Chaudron, Moufia, Les Camélias, Bas de la Rivière, and Primat), 2 in Sainte-Marie, and 2 in Sainte-Suzanne. In each one, a visit removes every breeding site in the yard and fits a cover (a lid or mosquito mesh) to stored water. It ends with one line on leptospirosis. The ARS's 2025 partner guide calls door-to-door clearance of yards the effective method, and expects communes and associations to do it. Its roughly 100 vector-control agents work around reported cases, so this round adds to the ARS's work rather than repeating it. Coordinate the targeting with the ARS. The Réunion drought framework order runs to 21 pages and never mentions mosquitoes. So CINOR's own water-cut notices should carry the line, under the island rule RE-01. The best trial evidence is Camino Verde (Andersson et al., *BMJ* 2015, 150 clusters, 85,182 people): community-led clearance cut recent dengue infection by about 25–30% and houses with larvae by 44%. Dengue circulation is sporadic now, with 347 cases on the island so far in 2026 (Santé publique France, 9 October). But the serotype mix has shifted: DENV2 made up 30% of typed cases by July, first detected in La Possession (Santé publique France, 10 July 2026). A second infection with a different serotype carries the higher risk of severe disease.

Corrections: covers are one item in the visit, not its core. Réunion field data (Delatte et al. 2008) found small discarded containers, not stored water, to be the main urban breeding site in the wet season, so the draft's two covers per household became one. The Kass'Moustik precedent (door-to-door since 2006) is documented but carries no measured effect size. **Transferability:** Camino Verde studied *Aedes aegypti* in Nicaragua and Mexico; Réunion's vector is *Ae. albopictus*, and no trial of it exists. The unit price (€10 a cover, retail-based, plus

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€3 delivery) has no Réunion quote. **Who is accountable:** the communes, as everywhere in the Réunion set; ARS vector-control work is already State-funded, so the ARS is consulted on targeting. **Delivery capacity:** no association with a delivery record at this scale is named yet, so the floor is a 1,000-household pilot, and the recipient is to be named and the money released against a written capacity check.

2. By 30 November, pre-position a debris crew and a generator at the Rivière Saint-Denis intake for every red alert, and sign a framework contract for drinking-water packs for the upper quarters

PARTIALLY CONFIRMED | PHASE: SAUVEGARDE | EMERGENCY MANAGER · ECONOMIST · HISTORIAN · FRONTLINE | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
CINOR, with SEMOP Dionéo and the city of Saint-Denis	**30 November 2026; crew in place at each alerte rouge**	**About €120,000 per event, released on the trigger (CINOR budget; estimate)**	**Several days without safe water in the hauts of Saint-Denis**

In Garance (28 February 2025), debris blocked the Rivière Saint-Denis intake, and the Bellepierre plant stopped producing. Dionéo had to clear the intake after the storm, and a pumping room flooded. On days 3 to 5, upper Bellepierre, Bellevue (whose main pipe broke), Le Brûlé, Saint-Bernard, La Source, and parts of La Montagne were still cut. The city sent tankers, but water packs reached only La Montagne and Saint-François. So the crew and the generator should be at the intake during the red alert, not called after the all-clear. Before the season, sign a framework contract with call-off orders (*accord-cadre à bons de commande*, Code de la commande publique R2162-2), which is the normal public-procurement tool for this. Name the distribution points now: La Montagne, Le Brûlé, Saint-Bernard, La Source, Bellepierre, and Bellevue. Trigger: alerte rouge, or any distribution unit cut for more than 24 hours. Cost arithmetic: 20,000 people × 3 L a day × 4 days × €0.50 a litre delivered = €120,000.

* Suggested formulation: check with a health worker before sharing.

Corrections: the plant stopped because its intake was blocked; neither source says the plant itself failed. La Providence and Saint-François were back by day 5, so La Source and Saint-Bernard replace them on the list. No source says Saint-Denis's Garance tankers carried non-potable water, so that claim is dropped (see *Do not* 5 for the drought case). €0.30 a litre was too low: the nearest real price is Mayotte, 2023, at €0.46 a litre retail and not delivered, so €0.50 is an estimate. Whether CINOR already holds such a contract is unknown.

3. By 15 November, list where people sleep rough in Saint-Denis, and have a known face offer shelter 48 hours before each orange alert

PARTIALLY CONFIRMED | PHASE: AVANT | FRONTLINE · EMERGENCY MANAGER · SKEPTIC · PHILANTHROPIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
CCAS of Saint-Denis, with the Croix-Rouge (Samu social, 115) and the Maison de la Fraternité	**List by 15 November 2026; rounds at each pré-alerte**	**€60,000 for lists, kits, and transport, not staff (F01)**	**Deaths of people sleeping in ravines and riverbeds during a cyclone**

All three of Réunion's confirmed Belal deaths (January 2024) were homeless people: one at Saint-Gilles, one drowned beside his tent in a riverbed in Salazie, and one in a shack in Le Tampon. Réunion La 1ère reported that the Saint-Gilles man had refused shelter; the Interior Minister said others "auraient refusé" (conditional). None of them died in CINOR, but Saint-Denis is where the island's street services are concentrated. Réunion has about 1,600 homeless people, more than 900 of them sleeping rough (Assemblée nationale; the source gives no year). The services already exist: the city's Maison de la Fraternité runs a daytime round, and the Croix-Rouge runs the Samu social and evening rounds on Tuesdays and Fridays. What is missing is a list of sleeping places (ravines, riverbeds, under bridges) made before the season. Then someone people already know offers shelter, water, and a phone charge at pré-alerte, with transport, and shelter rules agreed in advance on pets and belongings. The outreach itself is done by the State-funded teams; the money buys what they lack. Cost: the sleeping-places list (€10,000: mapping with the teams, printing, and phones, an estimate) + transport to shelter at each pré-alerte (€25,000: about 50 vehicle-days with a driver at €500, an estimate) + 300 kits × €50 + a €10,000 reserve.

* Suggested formulation: check with a health worker before sharing.

Corrections: Belal's dead were outside CINOR. Shelter refusal is reported for the Saint-Gilles man only; the minister's "auraient refusé" is conditional. A fourth Belal death from generator carbon monoxide appears in some counts but not in the prefecture's. "SAMU social 974" is the Croix-Rouge's service, not a separate body. Whether Réunion shelters refuse pets is unverified, so treat the pet rule as a design choice, not a documented barrier. The draft paid two outreach posts through the Croix-Rouge's Samu social, which the State already funds, so the line now pays for lists, kits, and transport for the existing teams instead of staff.

4. By 30 November, enrol isolated older people in each mayor's register of vulnerable people, and give each one a named neighbour who checks on them at alerte rouge, on day 2 of a water cut, and on hot days

PARTIALLY CONFIRMED | PHASE: AVANT | PUBLIC HEALTH · FRONTLINE · PHILANTHROPIST

ADAPTATION INVENTORY: HEAT WARNING SYSTEMS (ITALY, US, CANADA), NOT IN A SETTING LIKE THIS

OWNER	BY WHEN	COST	PREVENTS
The three mayors, through their CCAS, with neighbourhood associations	30 November 2026	€52,500: three coordinators for 5 months (F02)	Deaths of older people with chronic illness during heat and water cuts

The 45 chikungunya deaths of 2024–25 were mostly older people with several chronic illnesses (Santé publique France, 10 December 2025). 13.6% of Réunion adults have diagnosed diabetes, against 5.7% in mainland France (SPF, 2021). That is the comorbidity that turns dehydration during a water cut, or a hot night, into a death. Each mayor keeps a voluntary register of vulnerable people (Code de l'action sociale et des familles, L121-6-1) for the departmental emergency plan, which covers exceptional risks, not just heatwaves. Start with the 16 priority neighbourhoods, enrol people who agree, and pair each with a neighbour. There is no Météo-France heat warning in Réunion, so the plan has to name its own hot-day trigger. The Gillot seasonal count (19 days above 31°C expected) gives a starting point.

* Suggested formulation: check with a health worker before sharing.

Corrections: the register covers people aged 65 and over, people over 60 unable to work, and disabled adults, and only on request. People under 65 with diabetes or heart disease are not eligible, so keep them on a separate consented outreach list. The cost-effectiveness basis is Philadelphia's heat warning system (Teisberg, Ebi and others, 2004: 117 lives over 1995–98, about $210,000). **Transferability is weak:** it was a temperate city triggered by air-mass forecasts, not a humid tropical island.

5. By 25 November, sign CINOR's intercommunal safeguard plan with this water, heat, and epidemics annex, naming its triggers and owners

PARTIALLY CONFIRMED | PHASE: AVANT | EMERGENCY MANAGER · ECONOMIST · HISTORIAN | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
The president of CINOR, with the three mayors	25 November 2026 (legal deadline 26 November, AMF)	Staff time; no new money	Actions 1–4 and 7 staying voluntary when the alert comes

Under the loi Matras (2021-1520 of 25 November 2021) and décret 2022-907, intercommunalities like CINOR must have an intercommunal safeguard plan (PICS) within five years: by 25 November 2026 (legal deadline 26 November, AMF); co-adopted by the EPCI president and each mayor with a PCS. Writing this annex into it turns actions 1–4 and 7 from good intentions into assigned duties, with their triggers: the VigiEau level, alerte rouge, and a distribution unit cut for more than 24 hours.

Corrections: the plan is adopted by arrêté, not by a vote, and not by the president alone: each mayor with a communal safeguard plan (PCS) co-adopts it. The legal deadline is 26 November (AMF); aim for 25 November. No public source shows whether CINOR already has a PICS; ask CINOR.

6. By 30 November, prune trees over homes and power lines in Saint-Denis, signpost and close fords and ravine crossings at alerte orange, and broadcast the generator carbon-monoxide warning

PARTIALLY CONFIRMED | PHASE: AVANT | EMERGENCY MANAGER · HISTORIAN | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
The three communes (Saint-Denis first), with the municipal police	30 November 2026	Existing maintenance budgets	The kinds of death Garance caused in Saint-Denis

* Suggested formulation: check with a health worker before sharing.

Garance killed five people on Réunion, three of them in Saint-Denis: a woman swept away by water, a man in an electrical fire, and a man pinned under a tree. It is the only event in the record with confirmed deaths in CINOR. Pruning trees over homes and lines, closing fords and ravine crossings, and warning about generators indoors each address a cause of death on Réunion's own record. After cyclone Batsirai (February 2022), about ten people were poisoned by carbon monoxide (AFP).*

Corrections: no source says the woman died at a ford, so this briefing does not claim closures would have saved her. No Saint-Denis order closing fords at orange alert was found; the prefecture's orange-alert orders cover coastal access.

7. By 30 November, stock 3,000 water-storage kits and 2,000 no-cook food kits, released at pré-alerte or alerte orange before a forecast or announced cut, and have the three CCAS ready their own aid lists for the island trigger-cash line

PARTIALLY CONFIRMED | PHASE: PENDANT | FOOD SUPPLY · PHILANTHROPIST · ECONOMIST

ADAPTATION INVENTORY: ANTICIPATORY CASH (BANGLADESH, NEPAL) FOR THE CASH PART ONLY

OWNER	BY WHEN	COST	PREVENTS
The three CCAS, with the Croix-Rouge / PIROI (water kits) and the Banque alimentaire des Mascareignes (food)	30 November 2026; released at pré-alerte or alerte orange	€307,000 (target): €255,000 water kits (F04) + €52,000 food kits (F05); cash from the island line RE-F01	Households with rice and no water to cook it, and unsafe stored water

The food lens's finding: food is not what runs out on Réunion; water and power are. After Belal, the port worked again within three days, but households cut off from water cannot cook what they have. A water kit holds jerricans (two 20-litre food-grade jerricans with taps), purification tablets (60 one-litre Aquatabs), and a container cover. Kits go out at pré-alerte or alerte orange, before a forecast or announced cut: never at alerte rouge, which is lockdown, and not on a cut of more than 72 hours, which is when the State's bottled water starts (ARS Garance review). That is €64.47 in goods at mainland prices, plus about €20 for freight and distribution (an estimate), so €85. It is a storage-and-treatment kit, not drinking-water supply: after Belal the state sent 1.1 million bottles, about 700,000 of them through the CCAS and schools. A food kit covers three days for a household, at €26 (food-aid users' €3.70 a person a day × 2.35 people × 3 days, an estimate). The Croix-Rouge platform PIROI holds jerricans in its Réunion warehouse. The Banque alimentaire works through 11 CCAS and 30 associations. The three CCAS should also put together now the documents the State's emergency relief will need after the event:

* Suggested formulation: check with a health worker before sharing.

bank details, ID, income proof, and a list of uninsured households. These are the CCAS's own aid lists, kept separate from the mayor's register of vulnerable people, which Décret 2026-590 forbids using for cash. Then the island trigger cash (€475 a household, line RE-F01) can be paid within seven days rather than nine weeks.

Corrections: the Banque alimentaire handles food only, so the water kits go through the Croix-Rouge / PIROI. The State's emergency relief (*secours d'extrême urgence*) is opened by the préfet after the event, so dossiers can only be pre-assembled, not pre-filed. After Garance, nine weeks after landfall, €335,000 of the €574,000 committed had been paid.

8. After every cyclone or heavy-rain episode, the ARS issues a leptospirosis alert within 72 hours, and commune clean-up crews are offered the free vaccine before the season

PARTIALLY CONFIRMED | PHASE: SAUVEGARDE | PUBLIC HEALTH · HISTORIAN · SKEPTIC · FRONTLINE | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
ARS La Réunion, with the three communes for their crews	**Message ready by 30 November 2026; crews offered the vaccine from November**	**Within ARS budgets (the ARS and CGSS vaccine campaign is free)**	**Leptospirosis deaths after rain returns**

The message: boots and gloves, cover cuts, no bathing in fresh water, and anyone with a fever after exposure sees a doctor within 48 hours and asks for the PCR test. Réunion had 250 cases and 3 deaths this year to 9 October (Santé publique France); 89% of cases are men, 61% are hospitalised, and 28% go to critical care (Santé publique France, 17 April 2026). Météo-France expects rain above normal, and a post-cyclone rise was documented in 2014. Since 29 September 2025 the vaccine has been free for all adults (ARS and CGSS), with the most exposed prioritised, so commune clean-up crews come first: two injections 15 days apart, a booster at 4–6 months, then every 2 years if exposure continues. It is given at CHU Nord, CHU Sud, and occupational health services, so crews go through their occupational health service, or the ARS would need to run mobile sessions. Protection builds over the first month after two doses; it is not this season's main defence. So the vaccine sits alongside the boots, gloves, no wading, and early care in the message, not ahead of them (RE-X08). Do not offer doxycycline as routine prevention: the Cochrane review (CD007342) found its benefit unclear, and the ARS gives no such guidance.*

* Suggested formulation: check with a health worker before sharing.

Corrections: the North is Réunion's least-affected sector (16 of 142 cases by mid-April 2026, 7 per 100,000, against 34 per 100,000 in the East, the highest, and 18 per 100,000 in the South), so this ranks below the panel's placing. "96% men" was wrong: 89% of all cases, and 95% of hospitalised cases. After Belal the ARS issued an alert; no spike was documented.

9. Watch the monthly fresh-food price index, and have the Préfecture ready a request for a temporary price decree if fresh prices jump after a cyclone

PARTIALLY CONFIRMED | PHASE: SAUVEGARDE | FOOD SUPPLY | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Préfecture de La Réunion (requests); the Government (decrees)	Draft ready by 30 November 2026	Staff time	A Garance-sized fresh-price spike passing unchallenged

After Garance, fresh-product prices rose 19.1% in a month (INSEE, March 2025) and took until August to recover. In August 2026 they were up 0.4% on the month, so the drought has not reached prices yet. The Code de commerce (L410-2, third paragraph) allows the Government to regulate prices by decree for up to six months in a crisis or public calamity; Mayotte got one after Chido (décret 2024-1182). Trigger: fresh products up 10% or more on the month in the INSEE Réunion index.

Corrections: the préfet cannot set prices alone; a government decree is needed. The index is published about four weeks late, so it guides aid volumes, not the first week after a storm. The 2025 overseas cost-of-living bill lost its price-regulation clause in the Senate, and its final adoption is unverified; do not rely on it. Lives saved are indirect and small.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	ARS	CINOR	Water operators	Communes	CCAS	Croix-Rouge	Associations	Préfecture
1. Breeding-site and cover rounds	C	C	I	A/R	C	I	R	I
2. Intake crew and water packs	C	A	R	R	I	C		C
3. Rough-sleeper outreach	I	I		R	A/R	R	R	C
4. Vulnerable-people register	C			A	R	I	R	
5. Sign the PICS annex	C	A/R	C	R	I			C
6. Trees, fords, generators		I		A/R			I	C
7. Water and food kits; cash lists		C		I	A/R	R	R	I
8. Leptospirosis alert and vaccine	A/R			R	I		I	
9. Fresh-price watch				I			C	A/R

What the chart shows: the communes appear in every row, do the work in six, and are accountable for three, including the cover rounds (1), the largest donor line; the mayors co-adopt the PICS (5). The CCAS are accountable for the other two rows that carry donor money (3 and 7). CINOR is accountable for only two rows, but they are the two that need a decision rather than money: signing the PICS (5) and contracting the water packs (2). The ARS is accountable for the leptospirosis row (8) and consulted on the cover rounds, whose vector-control work is already State-funded. No row depends on the State paying first.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs.

FLOOR	TARGET	CEILING	LARGEST UNCERTAINTY
€279,000 (about $312,000)	€627,500 (about $702,000)	€1,138,600 (about $1,275,000)	How many households store water uncovered: no survey exists

At the ECB reference rate of 14 October 2026, €1 = US$1.1195, fixed for all six Réunion briefings. **Floor** = outreach lists, kits, and transport €60,000 + coordinators €52,500 + a 1,000-household covers pilot (€13,000) + 1,500 water kits (€127,500) + 1,000 food kits (€26,000) = €279,000. The covers line scales to the target and ceiling only once a recipient with a delivery record is named, against a written capacity check. **Target** = the same, with covers for 16,000 households (€208,000), 3,000 water kits (€255,000), and 2,000 food kits (€52,000) = €627,500. **Ceiling** = covers for all 31,700 poor households (92,833 households × 34.2%, an estimate, since the rate is per person; €412,100),

* Suggested formulation: check with a health worker before sharing.

6,000 water kits (€510,000), and 4,000 food kits (€104,000), plus the two fixed lines = €1,138,600. Above that, delivery capacity binds before the peak, not money. **Set unit costs for all six Réunion zones:** a cover €10 (retail drum nets €4.80–9.52 in mainland France, plus freight; a material-only estimate gives €6), delivered €13; a water-storage kit €85. **Not counted here:** the island trigger-cash line RE-F01 (€475 a household; €475,000 / €950,000 / €1,623,075 for all of Réunion), and the potable-water contract in action 2 (€120,000 per event), which is CINOR's own budget. **The target fell 55%** from the panel's €1,380,500: the cash (€475,000) moved to the island line, and covers went from two per household to one. **Already funded, so not duplicated:** the ARS vector-control campaign (€15M in early 2025) and the State's 65% share of about 400 mosquito-control jobs. Neither buys covers or kits.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
CCAS of Saint-Denis, for the outreach run by the State-funded Samu social (Croix-Rouge) and the city's Maison de la Fraternité	Action 3: sleeping-places list, kits, and transport to shelter at pré-alerte	€60,000 ($67,170)	Direct grant; readiness	The only cyclone deaths on the island in 2024 were rough sleepers. No published cost-per-life figure exists; the amount is the list €10,000 + transport €25,000 + 300 kits €15,000 + a €10,000 reserve (estimates). It pays for no staff, because the State funds the Samu social	PARTIALLY CONFIRMED
The CCAS of Saint-Denis, Sainte-Marie, and Sainte-Suzanne	Action 4: three register-and-neighbour coordinators	€52,500 ($58,774)	Top-up to an existing scheme; readiness	Philadelphia heat warnings, 117 lives over 1995–98 for about $210,000 (Teisberg and others, 2004); a temperate US city, weak transfer	PARTIALLY CONFIRMED
Recipient to be named; released against a written capacity check (neighbourhood associations under the ARS partner guide are the candidates)	Action 1: covers in the rounds; floor a 1,000-household pilot through one body with a delivery record, target 16,000, ceiling 31,700 households	€13,000 / €208,000 / €412,100 (floor / target / ceiling)	Pre-positioned stock; readiness	Girardot, Colombia: a covers programme at US$20.9 a household, cost-effective only where dengue is common (Taborda and others, 2022); our €13 is the goods only	PARTIALLY CONFIRMED

* Suggested formulation: check with a health worker before sharing.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Croix-Rouge / PIROI (Réunion warehouse)	Action 7: water-storage kits, 1,500 / 3,000 / 6,000	€127,500 / €255,000 / €510,000 (floor / target / ceiling)	Pre-positioned stock; released on the trigger	€64.47 in goods at mainland retail + about €20 freight and distribution (estimate); no cost-per-life figure	PARTIALLY CONFIRMED
Banque alimentaire des Mascareignes, through the three CCAS	Action 7: three-day no-cook food kits, 1,000 / 2,000 / 4,000	€26,000 / €52,000 / €104,000 (floor / target / ceiling)	Pre-positioned stock; released on the trigger	€3.70 a person a day (BAM, 2022) × 2.35 × 3 days = €26 (estimate)	PARTIALLY CONFIRMED

The funder's one thing: after Belal, the State's fund for uninsured people paid 56 people about €29,000 in total, about six months later (11 July 2024). After Garance, the CCAS handled 3,759 files and paid 3,417 households €1.6 million. The channel that works is the CCAS, so write its trigger rules before November (a board resolution on *aide facultative*), not after the alert. Goods shipped to an island after an alerte rouge arrive after the deaths.

* Suggested formulation: check with a health worker before sharing.

Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for this hazard and would cost lives here. Cards 1–6 apply to every Réunion zone and are written once for the island (RE-X01 to RE-X05, and RE-X07 for card 6); card 7 is specific to the North.

1. Do not call this season an "El Niño drought"

CONTRAINDICATED ISLAND-WIDE (RE-X01)

Météo-France expects El Niño to bring Réunion rain that is on average in surplus. The shortage is a carried-over deficit and leaking networks. A drought message tells people to expect less rain, when the risks are heavy rain on dry ground, flash floods, leptospirosis, and mosquitoes breeding in stored water.

Instead: use the premise: a record carried-over deficit and leaking networks, in a hot El Niño year in which rain, when it comes, brings its own dangers.

2. Do not make insecticide fogging the visible response to dengue or chikungunya

CONTRAINDICATED ISLAND-WIDE (RE-X02)

Deltamethrin has been used on Réunion since 2006, and resistance in *Ae. albopictus* was detected in 2017 (a 2025 preprint, not yet peer-reviewed, describes the mechanism). The ARS already limits spraying to the area around cases "to avoid creating resistance". Fogging streets looks like action and leaves the breeding sites in place.

* Suggested formulation: check with a health worker before sharing.

Instead: keep spraying to the ARS's case-centred use, and put the visible effort into door-to-door clearance (action 1).

3. Do not tell households to "store water" without saying "covered, emptied and scrubbed weekly"

`CONTRAINDICATED` `ISLAND-WIDE (RE-X03)`

Each rationing notice makes households fill tanks, buckets, and drums: the containers mosquitoes breed in, just as the hot, wet season starts. In Girardot, Colombia, people kept storing water after supply returned, and stored water was found in 95% of homes. That evidence is about *Ae. aegypti*, not Réunion's vector. The ARS's own campaign line is to cover water reserves with mesh or cloth.

Instead: "Couvrez vos réserves (couvercle ou moustiquaire), videz-les et frottez-les chaque semaine", on every water-cut notice (RE-01).

4. Do not push chikungunya vaccination with Ixchiq for people aged 65 and over, or put chikungunya vaccine stock ahead of dengue and leptospirosis

`CONTRAINDICATED` `ISLAND-WIDE (RE-X04)`

The HAS advised suspending Ixchiq for people aged 65 and over across France on 25 April 2025, after serious adverse events in older people, most of them on Réunion. That advice still stands; since June–July 2026 the EU has also restricted Ixchiq to people at high risk of infection. The HAS estimates that 66% of Réunion's population now carries chikungunya antibodies and judges a new epidemic "peu probable". Only 58 local cases have been recorded in 2026.

Instead: follow the HAS (24 April 2026): Vimkunya for people aged 65 and over and for 12–64-year-olds with comorbidities, only if an epidemic resumes. Spend the effort on dengue and leptospirosis.

5. Do not let tanker water be taken as drinking water

`CONTRAINDICATED` `ISLAND-WIDE (RE-X05)`

In the January 2025 drought, the ARS recorded 8 distribution units on tankers marked "non alimentaire" (not for drinking or cooking), serving 36,240 people, alongside bottled water for 28,240. Sainte-Marie was one of the five communes affected. After Garance, even restored tap water was to be boiled or not drunk for 48–72 hours.

* Suggested formulation: check with a health worker before sharing.

Instead: label every tanker "non potable" in French and Creole, and hand out drinking-water packs at the same point (action 2).*

6. Do not open the shelters and wait

CONTRAINDICATED ISLAND-WIDE (RE-X07)

All three of Belal's dead were homeless. Réunion La 1ère reported that the Saint-Gilles man had refused shelter; the Interior Minister said others "auraient refusé" (conditional). People sleeping rough do not answer a public announcement; they answer someone they know.

Instead: action 3: a list made before the season, a known face at pré-alerte, transport, and shelter rules agreed in advance.

7. Do not give heat advice that needs electricity (fans, air conditioning) for the days after a cyclone

CONTRAINDICATED NORTH (CINOR)

Bejisa (January 2014) cut power to 172,000–181,000 households, about half the island. The hottest days of the season come in January and February, the same months as the cyclones. This card is reasoning: no heat death on Réunion is on record.

Instead: shade, wet cloths, the cool hours of the day, water set aside in covered containers, and a neighbour's check (action 4).*

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
1948 (Jan)	Cyclone; the island's deadliest	165 dead, island-wide (no ENSO record before 1950)	—
1962 (Feb)	Cyclone Jenny; 250 km/h gusts at Gillot, Sainte-Marie (neutral, ONI −0.2)	36 dead island-wide, including 9 fishermen at sea	—
1977–78	Dengue (DENV-2) epidemic (weak El Niño, +0.7)	About 30% of the population infected	—

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
1980 (Jan)	Cyclone Hyacinthe, record rain (3,240 mm in 3 days at Commerson crater); Saint-Denis damaged (neutral: ONI at or above +0.5 for only three seasons, short of NOAA's five)	25 dead, 7,000–8,000 homeless, island-wide; none recorded in Saint-Denis	A leptospirosis rise followed
1989 (29 Jan)	Cyclone Firinga; Saint-Denis flooded, landslide in Sainte-Marie (La Niña, −1.6)	4 dead, island-wide	—
2002 (Jan)	Cyclone Dina; a wall collapsed in Saint-Denis (neutral, −0.1)	No direct deaths, 2 seriously injured, 5 to 6 indirect deaths; about a month later, 2 young people died on a debris dam it left in the Rivière des Pluies (IGE, 2003)	—
2005–06	Chikungunya epidemic (La Niña, −0.9)	203 death certificates	Door-to-door Kass'Moustik campaigns from 2006
2014 (Jan)	Cyclone Bejisa (neutral, −0.2)	1 dead, 17 injured; power cut to 172,000–181,000 households, water to about 40% of residents	A post-cyclone leptospirosis rise was clearly documented
2024 (Jan)	Cyclone Belal, the first violet alert (El Niño, +1.8)	3 homeless people dead, none in CINOR; a 4th death (carbon monoxide) is disputed	The State's fund paid 56 uninsured people about €29,000, about six months later (11 July 2024)
2024–25	Chikungunya epidemic (cool-neutral, −0.5)	About 54,500 confirmed cases, 45 deaths (22 directly caused)	Ixchiq advised against for people aged 65 and over
2025 (Feb)	Cyclone Garance (cool-neutral, −0.5)	5 dead, 3 of them in Saint-Denis; more than 310,000 without drinking water; €379M insured losses	EU Solidarity Fund, €21.2M, 8.5 months later; CCAS paid 3,417 households

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
2025–26	Driest rainy season in 54 years of records (−45%, provisional; −60 to −70% in the hauts), after −30% in 2024–25 (Météo-France; cool-neutral)	No deaths recorded; restrictions from the order of 9 Oct 2026	Saint-Denis at alerte renforcée

The recurrence nobody has noticed: the cyclone that killed 165 people taught Réunion to fear wind, and it built for wind. Since then, the deaths have moved to the days after the storm: no tap water, stored water breeding mosquitoes, and floodwater carrying leptospirosis. One epidemic (2005–06, 203 death certificates) killed more people than every cyclone since 1980 put together. And the North's worst seasons do not follow El Niño: of the events above, only Belal and one weak-El Niño season (the 1977–78 dengue epidemic) fell in El Niño years.

WHERE THE EXPERTS DISAGREE

UNRESOLVED

- **Chikungunya is the threat to prepare for** versus **dengue and leptospirosis are**. Last year's epidemic and the vaccine debate pull attention to chikungunya; the public health lens and the HAS (66% seroprevalence, a new epidemic "peu probable") point to dengue, sporadic now but with DENV2 at 30% of typed cases by July (Santé publique France, 10 July 2026). A rise in local chikungunya cases in Santé publique France's bulletins would settle it.
- **Stored water is the main breeding site during rationing** (Skeptic, Economist, Philanthropist) versus **small discarded containers are** (Delatte et al. 2008, the only Réunion field study). The Réunion data predate rationing on this scale. A larval survey of a rationed quarter in Saint-Denis this November would settle it, and would size the cover line.
- **Supply the whole zone after a cyclone** (Economist: about €196,000 a day for 218,278 people) versus **supply the named quarters that lose water** (Emergency manager: about €120,000 per event). This briefing takes the second, because Garance's cuts concentrated in the hauts of Saint-Denis. A wider outage in a stronger storm would make the first right.
- **A unit cost of €10 a cover** (retail-based) versus **€6** (material-based). The set uses €10 until a Réunion supplier quotes.

WHAT NOBODY COVERED

BLIND SPOT

No lens examined how much of the North's water depends on electricity: pumping stations, treatment plants, and the household pumps and tanks that the frontline lens found are common ("dans beaucoup de foyers, l'eau est pompée dans des réservoirs"). Nor did any lens examine EDF's restoration order after a cyclone. In Bejisa, power went to about half the island. If water returns only when power does, then actions 2

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and 7 are sized to the wrong outage. The same gap hides people on home dialysis or oxygen, whom a diabetes rate of 13.6% makes more numerous than on the mainland. Their count and their place in EDF's priority list should go into the PICS annex.

HOW TO READ THIS METHOD

The panel is author-built. Eight simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are resolved at Climate Impact Lab impact-region scale; Réunion is one region of about 890,000 people. They are not neighbourhood figures. The El Niño projection covers *heat only*; drought, flood, fire, smoke, and crop failure are additional to it and are carried here by the historical record rather than by a published number. Here the study's map also publishes figures for Réunion's four arrondissements. The Saint-Denis arrondissement's 2 deaths is the study's own row. The four rows sum to 6 against an island total of 5, which is a rounding effect.

HOW THIS BRIEFING WAS PRODUCED METHODOLOGY

- **Scope.** The location was taken as given: North (CINOR), Réunion, the first zone of six in the Réunion set, aligned to the five intercommunal water authorities. Climate Impact Lab impact region, as the report records it: REU.Re7bdb3a99bec931b, decoded from the Impact Map's own boundary file. It covers the whole island, about 890,000 people; its display label was not confirmed. The study's map publishes a finer row for the arrondissement of Saint-Denis (GADM REU.2_1), which INSEE confirms is exactly the three CINOR communes. Sub-commune detail is used for hazard exposure and action design, not for death counts.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. The figure here is the 6-month field (`6mo_hotonly_total_mean`) of the study's map (Flourish visualisation 30298258, data of 18 September 2026). It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Water outages, dengue, and leptospirosis are carried by the historical record.
- **Expert panel.** Eight simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, philanthropist, and food supply chain analyst) each researched this place independently and in parallel, from web sources, to a fixed brief. A

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ninth agent resolved the exposure figure. Each worked from a shared pack of facts already verified for the Réunion set, and each was told to re-check any fact it used.

- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Seven verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). The financial action was also checked for availability in French law, a named payer, a trigger, and whether it already exists. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed. The exposure figure and every row of the record were checked at source.
- **Who does what.** The RACI chart is built from the owners named in the verified actions, with one accountable party per action. It is a proposal: no organisation named in it has been contacted or has agreed to a role.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Consulted. The download link leads to a registration form, so the full dataset (96 entries, 29 countries) was read from the data embedded in the inventory page. It holds no entry for France, Réunion, Mauritius, or any small island, and none for mosquito-borne disease. The nearest precedents are heat warning systems (Italy, the US, Canada) and anticipatory cash for floods (Bangladesh, Nepal), shown in the tags where they apply.
- **Health wording.** Passages where a wording error could hurt someone (generators and electrical hazards after a cyclone, leptospirosis and when to see a doctor, drinking water, and cooling in the heat) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and labels its estimates. One exchange rate (ECB, 14 October 2026) is used across the Réunion set. Every source is listed below with its verification status.

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- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. Several primary pages returned 403 (DEAL's insurers' damage table, Légifrance), and figures from them were taken from other sources that quote them. No Réunion price was found for covers, kits, or delivered water. Whether CINOR already has a PICS or a water-pack contract is unknown. No commune-level record of the 1982–83, 1997–98, or 2015–16 El Niño seasons was found.
- **Revised 2026-10-15 after the set audit (fixes applied:** the cover-rounds line links to RE-01 as a related national item and the communes are accountable for it; the covers floor is a 1,000-household pilot with the recipient to be named, so the floor falls from €370,000 to €279,000; the outreach line buys lists, kits, and transport, not State-funded staff; kits are released at pré-alerte or alerte orange; the cash lists are the CCAS's own, separate from the register; Do not 6 is the island card RE-X07 and Do not 5 carries the ARS as owner; island-wide events reference the island file and count only CINOR's deaths (Garance 3); placements store their ceiling; Hyacinthe is neutral; the Jenny, Firinga, Dina, and Bejisa tolls, the leptospirosis rates and vaccine, the Belal wording and its six-month payment, the PICS deadline and co-adoption, the carbon-monoxide poisonings (Batsirai, 2022), the dengue serotypes (SPF, 10 July 2026), and the deficit wording follow the set's settled values.)

References · status after verification in brackets

Exposure: Climate Impact Lab, 451,000 at Risk and its map sep-feb-heat-deaths [confirmed] · Impact Map boundary file [region decoded; label unconfirmed] · Adaptation Inventory [consulted]

Water: La 1ère, 3 Mar 2025 [confirmed] · Imaz Press, 4 Mar 2025 [confirmed] · Office de l'eau, Chroniques de l'eau 133 [confirmed] · Prefectoral restriction order, 9 Oct 2026 and VigiEau API [confirmed] · ARS drought sanitary point, 30 Jan 2025 [confirmed] · Drought framework order (draft text) [confirmed: no mosquito mention] · Mayotte bottled-water prices, 2023 [price proxy]

Mosquitoes and health: Andersson et al., BMJ 2015 (Camino Verde) [confirmed; other vector] · Delatte et al. 2008, Réunion breeding sites [confirmed] · Kass'Moustik letter, EID 2015 [precedent, no effect size] · ARS partner guide 2025 [confirmed] · ARS campaign 2024 [confirmed] · Deltamethrin resistance preprint, 2025 [not peer-reviewed] · Girardot water storage [confirmed; other vector] · Taborda et al. 2022, covers costing [confirmed] · SPF Réunion bulletin, 9 Oct 2026 [confirmed] · SPF Réunion bulletin, 10 Jul 2026 (DENV2) [audit-confirmed] · ARS leptospirosis vaccine leaflet [set value] ·SPF leptospirosis bulletin, 17 Apr 2026 [confirmed] · Cochrane CD007342 [confirmed] · HAS, Ixchiq, 25 Apr 2025 [confirmed] · Ixchiq EU restriction, 2026 [confirmed] · HAS avis 2026-0027 [confirmed] · SPF chikungunya deaths, 10 Dec 2025 [confirmed via summary] · SPF diabetes, 2021 [confirmed] · Teisberg et al. 2004 [confirmed; weak transfer]

People: Belal deaths, Outremers360 [confirmed] · Assemblée nationale, homelessness [confirmed] · QPV list 2024 [confirmed] · Garance deaths, ici and Brut [confirmed] · Batsirai CO poisonings (AFP, Feb 2022) [corrected: Batsirai, not Belal] · AMF, PICS deadline [set value] · INSEE, CINOR households and poverty [confirmed]

Money: Garance SEU payments [confirmed] · SEU payment speed [confirmed] · Belal FSOM [confirmed] · SEU circular, 2017 [confirmed] · EU Solidarity Fund, Garance [confirmed] · ARS vector-control spending, 2025 [confirmed] · State share of mosquito-control jobs [confirmed] · Jerrican price [€21.14, mainland] · Drum net price [mainland] · Banque alimentaire figures [confirmed] · PIROI warehouse [confirmed]

* Suggested formulation: check with a health worker before sharing.

Food and record: INSEE Réunion prices, April 2025 flash and August 2026 [confirmed] · Décret 2024-1182 (Mayotte prices) [confirmed] · DEAL EPRI events table [confirmed, cross-checked] · NOAA ONI v6 [confirmed] · SPF leptospirosis burden [confirmed for 1980] · Bejisa, The Local [confirmed] · Météo-France season review 2025–26, 29 Apr 2026 [set value] · IGE report on Dina, 2003 [set value] · Jenny, Firinga, and Bejisa tolls, fr.wikipedia [set values]

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